

University Physics Quiz 1

Name: _____ SID: _____ School/Department: _____

1. A particle moves outward along a spiral trajectory given by $r = A\theta$, where A is a positive constant. θ increases in time t according to $\theta = \alpha t^2/2$, where α is a positive constant.

(a) Find the time when the radial acceleration of the particle is zero.

(b) Find the time when the radial acceleration and the transverse acceleration of the particle have the equal magnitude.

Solution: From

$$\vec{r} = r\vec{e}_r \quad \dot{\vec{r}} = \dot{r}\vec{e}_r + r\dot{\theta}\vec{e}_\theta \quad \left(\frac{d\vec{e}_r}{dt} = \dot{\theta}\vec{e}_\theta, \frac{d\vec{e}_\theta}{dt} = -\dot{\theta}\vec{e}_r\right) \quad \ddot{\vec{r}} = (\ddot{r} - r\dot{\theta}^2)\vec{e}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta})\vec{e}_\theta,$$

we know that the radial acceleration and the transverse acceleration are respectively

$$a_r = \ddot{r} - r\dot{\theta}^2 \quad \text{and} \quad a_\theta = 2\dot{r}\dot{\theta} + r\ddot{\theta}.$$

Since $\theta = \frac{1}{2}\alpha t^2$ and $r = A\theta = \frac{A}{2}\alpha t^2$, we have

$$\dot{\theta} = \alpha t, \quad \ddot{\theta} = \alpha, \quad \dot{r} = A\alpha t \quad \text{and} \quad \ddot{r} = A\alpha.$$

Substituting above results into the formulae for the radial acceleration and the transverse acceleration, we have

$$a_r = A\alpha - \frac{1}{2}A\alpha^3 t^4, \quad a_\theta = 2A\alpha^2 t^2 + \frac{1}{2}A\alpha^2 t^2.$$

$$(a) \quad a_r = A\alpha - \frac{1}{2}A\alpha^3 t^4 = 0, \quad \alpha^2 t^4 = 2, \quad t = \frac{\sqrt[4]{2}}{\sqrt{\alpha}}.$$

$$(b) \quad a_r = A\alpha - \frac{1}{2}A\alpha^3 t^4 = a_\theta = 2A\alpha^2 t^2 + \frac{1}{2}A\alpha^2 t^2, \quad \alpha^2 t^4 + 5\alpha t^2 - 2 = 0, \quad \alpha t^2 = \frac{\sqrt{33}-5}{2},$$

$$t = \frac{\sqrt{\sqrt{33}-5}}{\sqrt{2\alpha}}.$$

2. A particle travels along the path $y = a + bx + cx^2$ on a plane, where a , b , c are constants. The speed of the particle is constant v_0 .

(a) Determine the x and y components of velocity at $x = 0$.

(b) Find the acceleration of the particle at $x = 0$.

Solution:

$$(a) \quad \dot{y} = b\dot{x} + 2cx\dot{x}.$$

$$\text{At } x = 0, \quad \dot{y} = b\dot{x}.$$

Substituting $\dot{y} = b\dot{x}$ into $\dot{x}^2 + \dot{y}^2 = v_0^2$, we have

$$(1 + b^2)\dot{x}^2 = v_0^2.$$

$$\text{Then, we have } v_x = \dot{x} = \frac{v_0}{\sqrt{1+b^2}}, v_y = \dot{y} = \frac{bv_0}{\sqrt{1+b^2}} \quad \text{or} \quad v_x = -\frac{v_0}{\sqrt{1+b^2}}, v_y = -\frac{bv_0}{\sqrt{1+b^2}}.$$

$$(b) \quad \ddot{y} = b\ddot{x} + 2c\dot{x}^2 + 2cx\ddot{x}$$

Making derivative on both side of $\dot{x}^2 + \dot{y}^2 = v_0^2$, we have

$$2\ddot{x}\dot{y} + 2\dot{y}\ddot{y} = 0, \quad \ddot{y} = -\frac{\dot{x}}{\dot{y}}\ddot{x}$$

$$\text{At } x=0, \quad \ddot{y} = -\frac{1}{b}\ddot{x}, \quad \ddot{y} = b\ddot{x} + 2c\dot{x}^2 \Rightarrow (1+b^2)\ddot{x} = -2bc\dot{x}^2$$

$$a_x = \ddot{x} = \frac{-2bc\dot{x}^2}{(1+b^2)} = \frac{-2bcv_0^2}{(1+b^2)^2},$$

$$a_y = \ddot{y} = -\frac{1}{b}\ddot{x} = \frac{2cv_0^2}{(1+b^2)^2}.$$

3. In spherical polar coordinates, the position vector of a particle is $\mathbf{r} = r\mathbf{e}_r$. If a particle moves in 3-dimensional space, its trajectory can be described with $r = r(t), \theta = \theta(t), \varphi = \varphi(t)$ in spherical polar coordinates, where t is the time. Evaluate the components of the velocity and acceleration of the particle in spherical polar coordinates.

Solution:

$$\mathbf{e}_r = \sin\theta(\cos\varphi\mathbf{i} + \sin\varphi\mathbf{j}) + \cos\theta\mathbf{k},$$

$$\mathbf{e}_\theta = \cos\theta(\cos\varphi\mathbf{i} + \sin\varphi\mathbf{j}) - \sin\theta\mathbf{k},$$

$$\mathbf{e}_\varphi = -\sin\varphi\mathbf{i} + \cos\varphi\mathbf{j}.$$

$$\dot{\mathbf{e}}_r = [\cos\theta(\cos\varphi\mathbf{i} + \sin\varphi\mathbf{j}) - \sin\theta\mathbf{k}]\dot{\theta} + \sin\theta(-\sin\varphi\mathbf{i} + \cos\varphi\mathbf{j})\dot{\varphi} = \dot{\theta}\mathbf{e}_\theta + \sin\theta\dot{\varphi}\mathbf{e}_\varphi,$$

$$\dot{\mathbf{e}}_\theta = [-\sin\theta(\cos\varphi\mathbf{i} + \sin\varphi\mathbf{j}) - \cos\theta\mathbf{k}]\dot{\theta} + \cos\theta(-\sin\varphi\mathbf{i} + \cos\varphi\mathbf{j})\dot{\varphi} = -\dot{\theta}\mathbf{e}_r + \cos\theta\dot{\varphi}\mathbf{e}_\varphi,$$

$$\dot{\mathbf{e}}_\varphi = -\sin\varphi\mathbf{i} + \cos\varphi\mathbf{j} = (-\cos\varphi\mathbf{i} - \sin\varphi\mathbf{j})\dot{\varphi} = -\dot{\varphi}(\sin\theta\mathbf{e}_r + \cos\theta\mathbf{e}_\theta).$$

$$\mathbf{r} = r\mathbf{e}_r, \quad \mathbf{v} = \dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta + r\sin\theta\dot{\varphi}\mathbf{e}_\varphi,$$

$$\mathbf{a} = \dot{\mathbf{v}} = \ddot{\mathbf{r}}$$

$$= \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta + \dot{r}\sin\theta\dot{\varphi}\mathbf{e}_\varphi + r\cos\theta\dot{\theta}\dot{\varphi}\mathbf{e}_\varphi + r\sin\theta\dot{\varphi}\dot{\mathbf{e}}_\varphi + r\sin\theta\dot{\varphi}\dot{\mathbf{e}}_\varphi$$

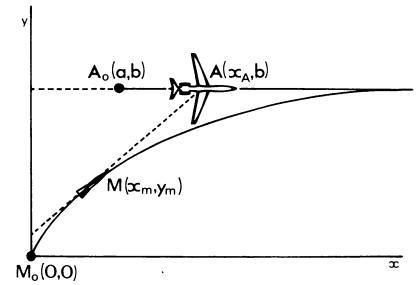
$$= \ddot{r}\mathbf{e}_r + \dot{r}(\dot{\theta}\mathbf{e}_\theta + \sin\theta\dot{\varphi}\mathbf{e}_\varphi) + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}(-\dot{\mathbf{e}}_r + \cos\theta\dot{\varphi}\mathbf{e}_\varphi)$$

$$+ \dot{r}\sin\theta\dot{\varphi}\mathbf{e}_\varphi + r\cos\theta\dot{\theta}\dot{\varphi}\mathbf{e}_\varphi + r\sin\theta\dot{\varphi}\dot{\mathbf{e}}_\varphi + r\sin\theta\dot{\varphi}[-\dot{\varphi}(\sin\theta\mathbf{e}_r + \cos\theta\mathbf{e}_\theta)]$$

$$= (\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\varphi}^2)\mathbf{e}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta} - r\sin\theta\cos\theta\dot{\varphi}^2)\mathbf{e}_\theta$$

$$+ (2\sin\theta\dot{r}\dot{\varphi} + 2r\cos\theta\dot{\theta}\dot{\varphi} + r\sin\theta\ddot{\varphi})\mathbf{e}_\varphi$$

4. (Pursuit curve) As shown in the figure, an aircraft (pursuee) is flying in a straight line parallel to the x-axis with speed v_A . A guided missile (pursuer) with constant speed v_M ($v_M > v_A$), moves towards the aircraft in such a way that the guidance of the missile locks on the aircraft and its direction of velocity always points to the aircraft. Initially, the aircraft is at point $A_0(a, b)$ and the missile is fired at the origin. Find the time used for the missile to hit the aircraft.



Solution: The position vector of the aircraft is $\mathbf{r}_A = (a + v_A t)\mathbf{i} + b\mathbf{j}$ and the position vector of the missile can be denoted as $\mathbf{r}_M = x(t)\mathbf{i} + y(t)\mathbf{j}$, where t is the time. Since the tangent vector of the trajectory of the missile always points to the aircraft, the equation of pursuit can be written as

$$\frac{\mathbf{r}_A - \mathbf{r}_M}{|\mathbf{r}_A - \mathbf{r}_M|} \cdot \frac{\dot{\mathbf{r}}_M}{|\dot{\mathbf{r}}_M|} = 1, \quad |\dot{\mathbf{r}}_M| = v_M. \quad (1)$$

The above equation can be rewritten as

$$\frac{(a + v_A t - x(t))\dot{x} + (b - y(t))\dot{y}}{\sqrt{(a + v_A t - x(t))^2 + (b - y(t))^2}} = v_M, \quad \dot{x}^2 + \dot{y}^2 = v_M^2. \quad (2)$$

Squaring and rearranging (2) gives

$$[(a + v_A t - x(t))\dot{x} + (b - y(t))\dot{y}]^2 = v_M^2 [(a + v_A t - x(t))^2 + (b - y(t))^2]$$

Expanding above equation, we obtain

$$(a + v_A t - x(t))^2 (\dot{x}^2 - v_M^2) + (b - y(t))^2 (\dot{y}^2 - v_M^2) + 2(a + v_A t - x(t))(b - y(t))\dot{x}\dot{y} = 0.$$

Using $\dot{x}^2 + \dot{y}^2 = v_M^2$, the above equation can be rewritten as

$$-(a + v_A t - x(t))^2 \dot{y}^2 - (b - y(t))^2 \dot{x}^2 + 2(a + v_A t - x(t))(b - y(t))\dot{x}\dot{y} = 0$$

or

$$(x(t) - a - v_A t)\dot{y} + (b - y(t))\dot{x} = 0. \quad (3)$$

((Another way to obtain (3):

The unit tangent vector of the curve is $\mathbf{t} = \frac{\dot{\mathbf{r}}}{|\dot{\mathbf{r}}|} = \frac{\dot{x}\mathbf{i} + \dot{y}\mathbf{j}}{\sqrt{\dot{x}^2 + \dot{y}^2}}$. Then, the unit normal vector is $\mathbf{n} = \frac{-\dot{y}\mathbf{i} + \dot{x}\mathbf{j}}{\sqrt{\dot{x}^2 + \dot{y}^2}}$. Thus,

the equation of pursuit can be written as

$$\mathbf{n} \cdot (\mathbf{r}_A - \mathbf{r}_M) = 0$$

or $(x(t) - a - v_A t)\dot{y} + (b - y(t))\dot{x} = 0. \quad)))$

By applying $x' = \frac{dx}{dy} = \frac{dx/dt}{dy/dt} = \frac{\dot{x}}{\dot{y}}$, we have

$$(x - a - v_A t) + (b - y)x' = 0. \quad (4)$$

Noting that the arc length is

$$s = \int_0^y \sqrt{1 + x'^2} dy = v_M t,$$

Eq.(4) can be rewritten as

$$\left(x - a - \frac{v_A}{v_M} \int_0^y \sqrt{1 + x'^2} dy \right) + (b - y)x' = 0. \quad (5)$$

Differentiating $\left(\frac{d}{dy} \right)$ above equation gives

$$-\frac{v_A}{v_M} \sqrt{1 + x'^2} + (b - y)x'' = 0 \quad \text{or} \quad \frac{d^2 x / dy^2}{\sqrt{1 + \left(\frac{dx}{dy} \right)^2}} = \frac{v_A / v_M}{b - y}.$$

Substituting $u = \frac{dx}{dy}$ into above equation, we have

$$\frac{du}{\sqrt{1 + u^2}} = \frac{v_A / v_M}{b - y} dy.$$

Noting that for $t = 0$, $y = 0$, $u = a/b$, integrating both sides of above equation, we have

$$\int_{a/b}^u \frac{du}{\sqrt{1+u^2}} = \int_0^y \frac{v_A/v_M}{b-y} dy \quad \text{or} \quad \ln \left(\frac{u + \sqrt{1+u^2}}{\frac{a}{b} + \sqrt{1+\frac{a^2}{b^2}}} \right) = -\frac{v_A}{v_M} \ln \left(\frac{b-y}{b} \right) \quad (b-y > 0 !!!)$$

or
$$\sqrt{1+u^2} = \left(\frac{b}{b-y} \right)^{\frac{v_A}{v_M}} \left(\frac{a}{b} + \sqrt{1+\frac{a^2}{b^2}} \right) - u.$$

Thus, the capture time is

$$\begin{aligned} t = \frac{s}{v_M} &= \frac{1}{v_M} \int_0^b \sqrt{1+u^2} dy - \frac{1}{v_M} \int_0^b \left(\frac{b}{b-y} \right)^{\frac{v_A}{v_M}} \left(\frac{a}{b} + \sqrt{1+\frac{a^2}{b^2}} \right) dy - \frac{1}{v_M} \int_0^b u dy \\ &= \frac{1}{v_M - v_A} \left[-b^{v_A/v_M} (b-y)^{\frac{v_M-v_A}{v_M}} \left(\frac{a}{b} + \sqrt{1+\frac{a^2}{b^2}} \right) \right]_0^b - \frac{1}{v_M} \int_0^b \frac{dx}{dy} dy \\ &= \frac{1}{v_M - v_A} \left(a + \sqrt{a^2 + b^2} \right) - \frac{1}{v_M} [x(y=b) - x(y=0)] \end{aligned} \quad (6)$$

Obviously, $x(y=b) - x(y=0) = a + v_A t$. Substituting that into (6), we have

$$t = \frac{1}{v_M - v_A} \left(a + \sqrt{a^2 + b^2} \right) - \frac{1}{v_M} (a + v_A t)$$

Finally, we obtain

$$t = \frac{1}{v_M^2 - v_A^2} \left(v_A a + v_M \sqrt{a^2 + b^2} \right).$$

To obtain the analytical solution of the pursuit curve equation, please refer to:

<http://mathworld.wolfram.com/PursuitCurve.html>